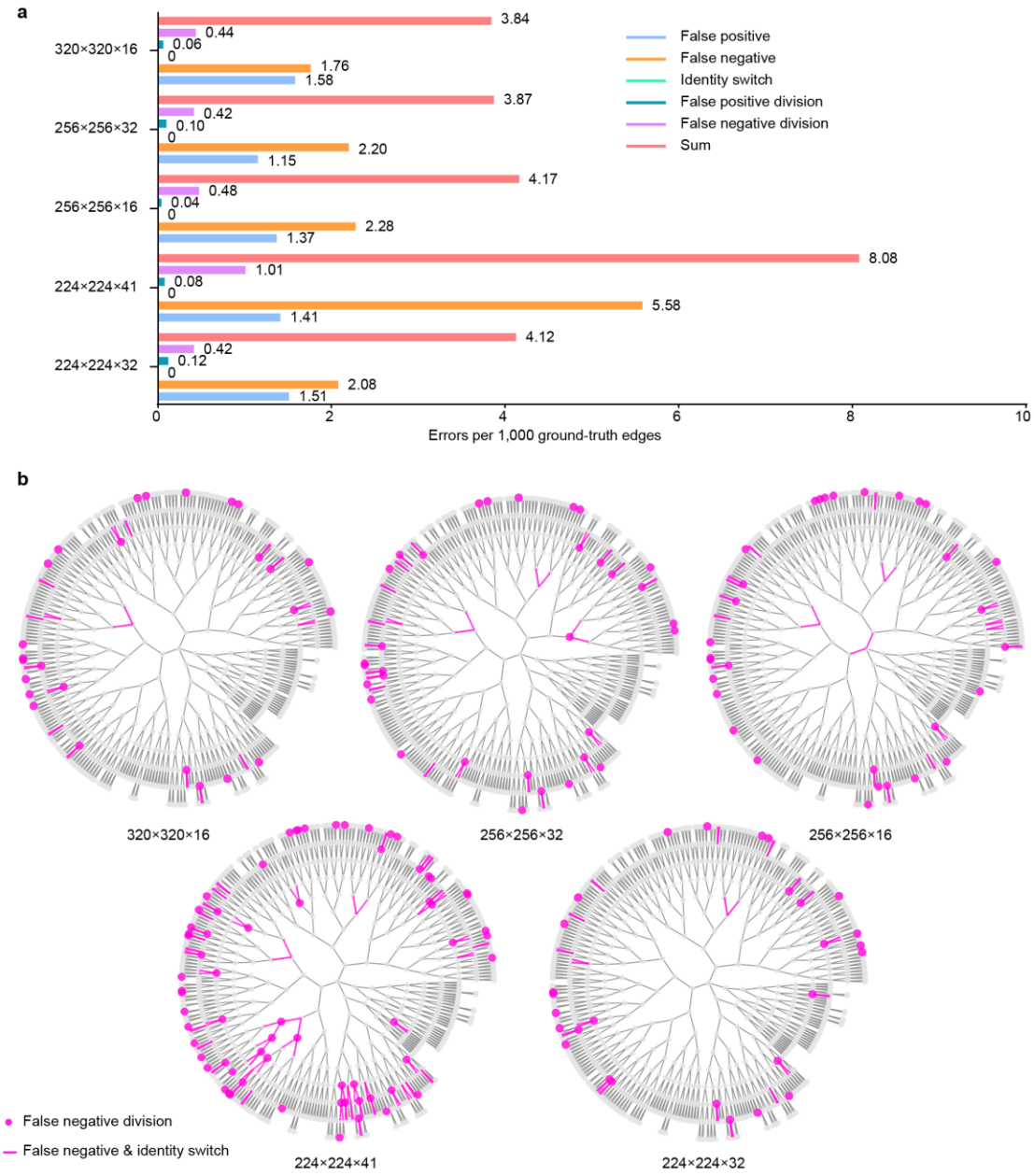


Supplementary Figure 9 | Influence of patch size on tracking performance. The impact of training size on the model's performance was evaluated, revealing that performance is primarily influenced by the training size and the axial resolution of the image. In general, larger training sizes lead to improved performance. However, for the 224×224×41 configuration (where the dataset contains 41 axial slices), the patch spans the entire z-axis, resulting in reduced variability in axial sampling during training and thus lower tracking accuracy. Error connections are highlighted in pink, while correct connections are depicted in grey. This data was retrained following the experiments presented in **Fig. 2**. All evaluation data in this figure were drawn from mskcc-confocal dataset #2. All models were trained using mskcc-confocal dataset #3.